

Chapter 12: Compare and contrast concepts and strategies to protect data

Overview

Protecting data is essential in modern cybersecurity. Data comes in various types and classifications, requiring different strategies to secure it across its life cycle—at rest, in transit, and in use. Organizations must align protection methods with the data's sensitivity, legal obligations, and business value.

1. Data Types

Different types of data must be protected due to sensitivity, regulations, or business importance:

- **PII (Personally Identifiable Information):** Data unique to an individual (e.g., SSNs, email, biometrics).
- **PHI (Protected Health Information):** Medical records and history, governed by laws like HIPAA.
- **Financial Data:** Includes banking, credit card, and transaction data; subject to financial regulations.
- **Legal Data:** Confidential legal documents and court records.
- **Intellectual Property (IP):** Patents, trade secrets, and copyrights.
- **Consumer/Customer Data:** Includes behavioral and transactional information, often regulated.
- **Government Data:** Controlled by strict rules regarding access, handling, and disposal.
- **Proprietary & Trade Secrets:** Internal business knowledge, R&D, and strategy data.
- **Biometric Data:** Includes fingerprints, facial recognition, etc.

2. Data Classifications

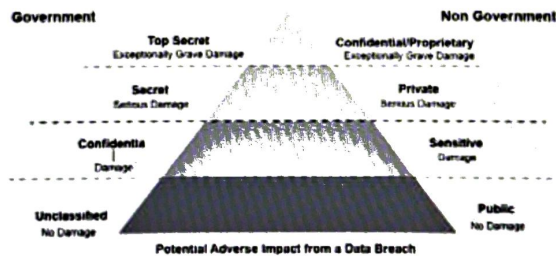


Figure 12.1: Governmental versus non-governmental data

Data is classified based on its sensitivity and impact if disclosed:

- **Public:** No confidentiality requirement (e.g., brochures).
- **Sensitive:** Could cause harm if exposed; often includes PII.
- **Private:** Intended to remain within a trusted circle; damaging if leaked.
- **Confidential:** Restricted access; e.g., R&D or legal documents.
- **Restricted:** Highly sensitive data with strict access and usage policies.
- **Critical:** Essential to operations (e.g., backups, encryption keys).

Classification guides storage, access control, and destruction policies.

4. General Data Considerations

- In information security, understanding data classification, states, and legal/geographical implications is essential for effective data management and protection.
- Data states define the context in which data resides and how it is accessed and utilized. There are three possible data states, described as follows:
 - **Data at Rest:** Stored data (e.g., databases, hard drives). Needs encryption and access control.
 - **Data in Transit:** Data being transmitted across networks. Protected by protocols like TLS, HTTPS.
 - **Data in Use:** Data being actively processed (e.g., in RAM). Requires memory-level protection.
 - **Data Sovereignty:** Data must comply with the laws of the country where it's stored or processed.
 - **Geolocation:** Helps verify access legitimacy by tracking physical origin of user requests.

5. Methods to Secure Data

Protecting data is vital for any organization, and various methods are used to ensure its confidentiality, integrity, and availability. Key techniques include:

- **Geographic Restrictions:** Limit data access based on physical location to comply with regional laws and improve security, though it may hinder remote or international operations.
- **Encryption:** Converts readable data (plaintext) into unreadable ciphertext, only accessible with a decryption key, providing strong protection against unauthorized access.
- **Hashing:** Transforms data into a fixed-length string through a one-way process, commonly used for password storage and ensuring data integrity, as hashes change if the original data is altered.
- **Masking:** Hides parts of sensitive data (e.g., displaying only the last four digits of a credit card number), allowing safe use in non-secure environments without exposing full information.
- **Tokenization:** Replaces sensitive data with unique, non-sensitive identifiers (tokens) that retain no exploitable value, often used in payment systems and identity protection.

These strategies work together to create layered, context-specific defenses tailored to an organization's data protection needs.